



From Synthetic Examples to Production Signals

Multimodal Training Data Pipelines with
Privacy-Safe Feedback

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AGENDA

Getting Setup

NeMo Data Designer

NeMo Anonymizer

Notebook 1: Text QA SDG with
NeMo Data Designer

Notebook 2: VQA SDG with
NeMo Data Designer

Notebook 3: Privacy with Anonymizer

Wrap-up



Getting Setup

<https://github.com/nabinchha/pydata-london-2026-data-designer-anonymizer>

Provider	Env var	Why	Sign-up
NVIDIA Build (default)	NVIDIA_API_KEY	Free key; hosts the VLM used in Notebook 2 and both Anonymizer model stages in Notebook 3	build.nvidia.com
OpenRouter	OPENROUTER_API_KEY	Hosted LLM provider. If selected in a notebook, LLM stages use OpenRouter; Notebook 3 entity detection uses the workshop Brev GLiNER endpoint	openrouter.ai
OpenAI	OPENAI_API_KEY	Hosted LLM provider. If selected in a notebook, LLM stages use OpenAI; Notebook 3 entity detection uses the workshop Brev GLiNER endpoint	platform.openai.com

Setup at least one of these to follow along

Getting Setup

<https://github.com/nabinchha/pydata-london-2026-data-designer-anonymizer>

Quickstart

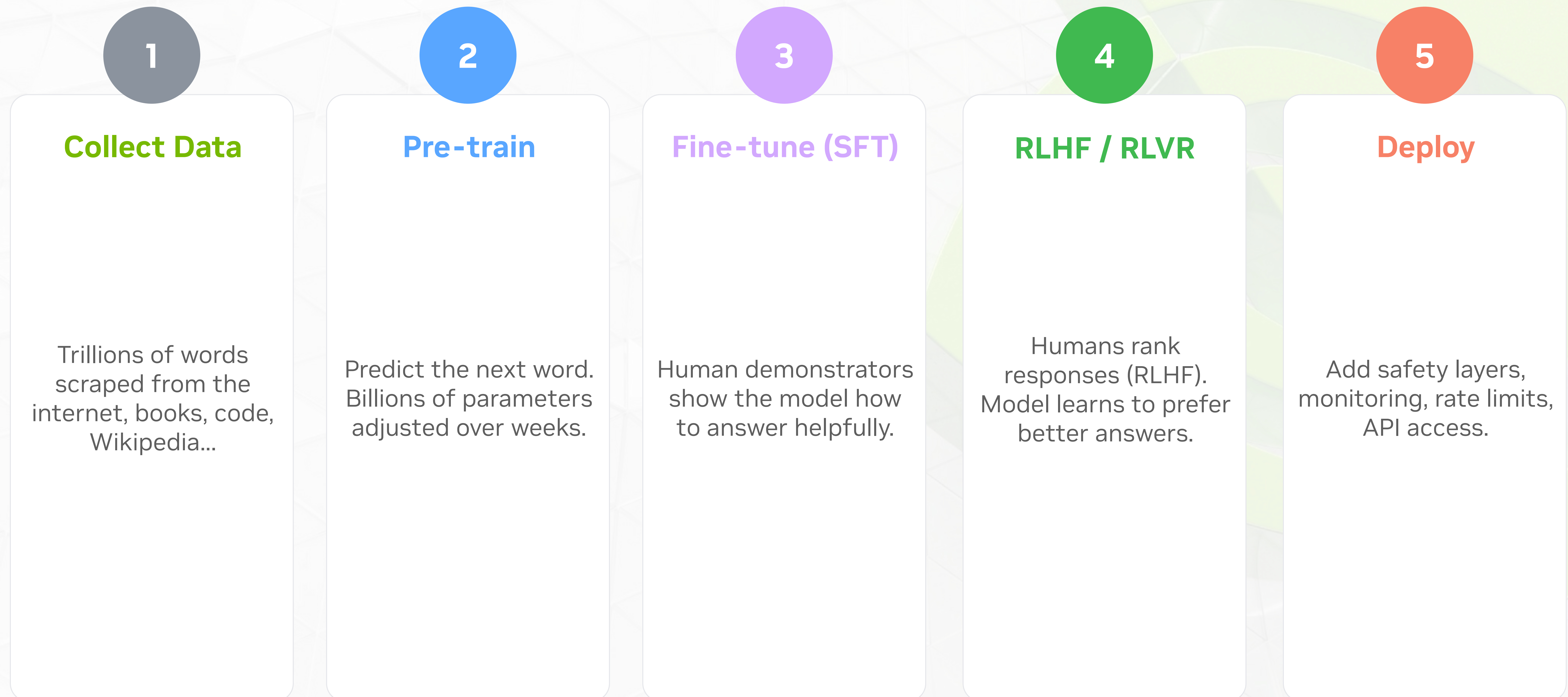
Run these commands to get set up before running the notebooks. Make sure **one** provider key is available, either in your shell environment or in `.env` : `NVIDIA_API_KEY` , `OPENROUTER_API_KEY` , or `OPENAI_API_KEY` .

```
git clone https://github.com/nabinchha/pydata-london-2026-data-designer-anonymizer.git
cd pydata-london-2026-data-designer-anonymizer
```

```
cp .env.example .env          # optional if your shell already exports a key
make setup                    # installs deps + runs offline checks
make lab                      # opens notebook UI
```

`make setup` runs `uv sync` , then `uv run python smoke_test.py` . The smoke test checks Python, imports, masked API key detection, a tiny local Data Designer pipeline, and the checked-in seed files. It does not make API

How are LLMs made?



NeMo Data Designer



NeMo Data Designer

What is it?

A **general framework** for generating high-quality synthetic data **from scratch** or based on **your own seed data**.

- **General Framework**
 - Use-case agnostic
 - Modular and extensible
- **From scratch**
 - Declarative config → dataset
- **You own seed data**
 - Domain-grounded data generation

Open Source Library:

<https://github.com/NVIDIA-NeMo/DataDesigner>

```
pip install data-designer
uv add data-designer
```

The screenshot shows the GitHub repository for NeMo Data Designer. The main content area displays the repository name, a description: "Generate high-quality synthetic datasets from scratch or using your own seed data.", and a "Welcome!" section. Below this, it states: "Data Designer helps you create synthetic datasets that go beyond simple LLM prompting. Whether you need diverse statistical distributions, meaningful correlations between fields, or validated high-quality outputs, Data Designer provides a flexible framework for building production-grade synthetic data." The "What can you do with Data Designer?" section lists three bullet points: "Generate diverse data using statistical samplers, LLMs, or existing seed datasets", "Control relationships between fields with dependency-aware generation", and "Validate quality with built-in Python, SQL, and custom local and". The right sidebar shows the "About" section with a description: "NeMo Data Designer: Generate high-quality synthetic data from scratch or from seed data.", a link to "nvidia-nemo.github.io/Data...", and a list of repository features like "Readme", "Apache-2.0 license", "Code of conduct", etc. The "Contributors" section shows 30 contributors.

NeMo Data Designer

Usage snapshot ~ 01/01-2026 - 06/04/2026

2.1T+

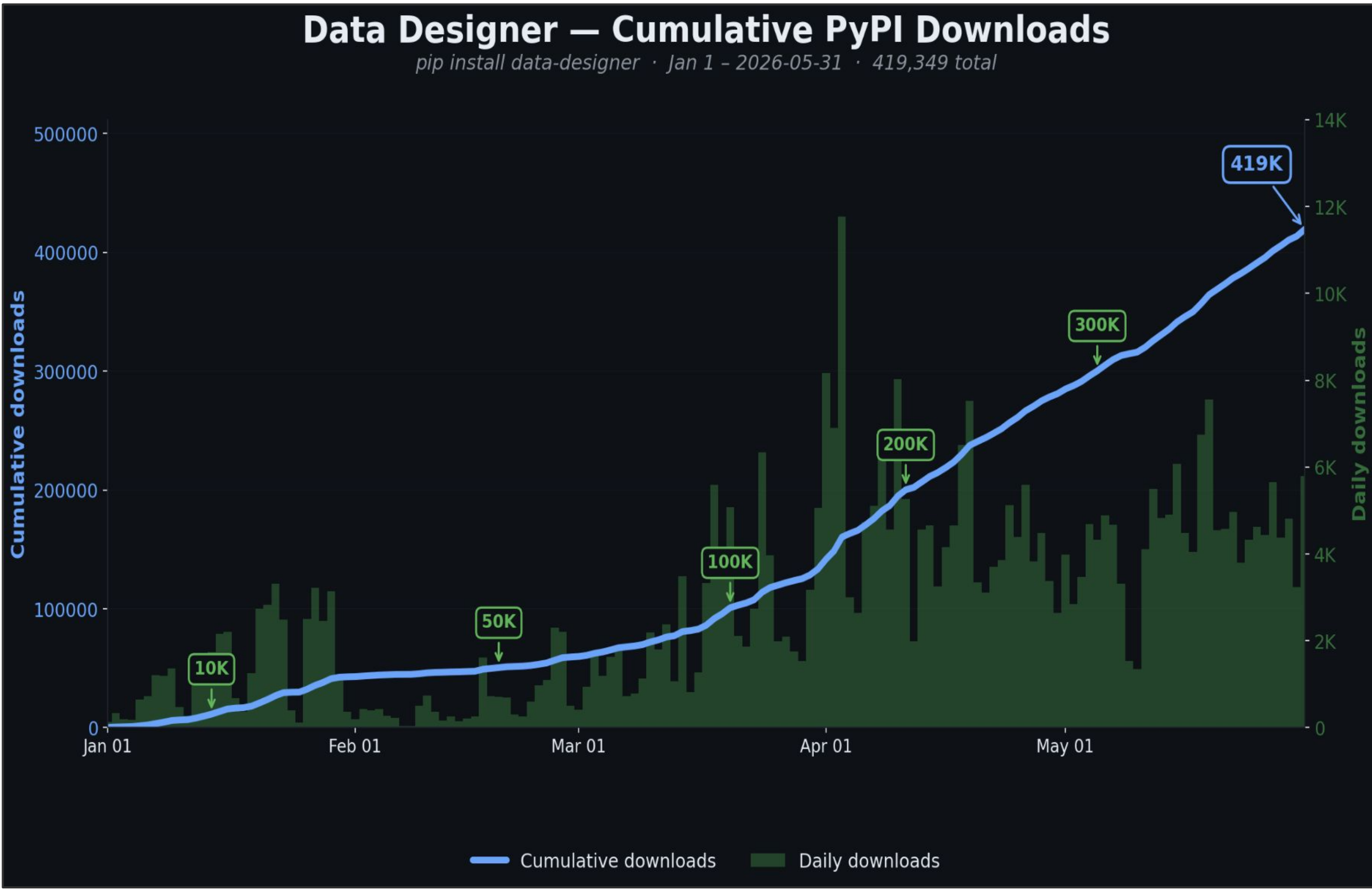
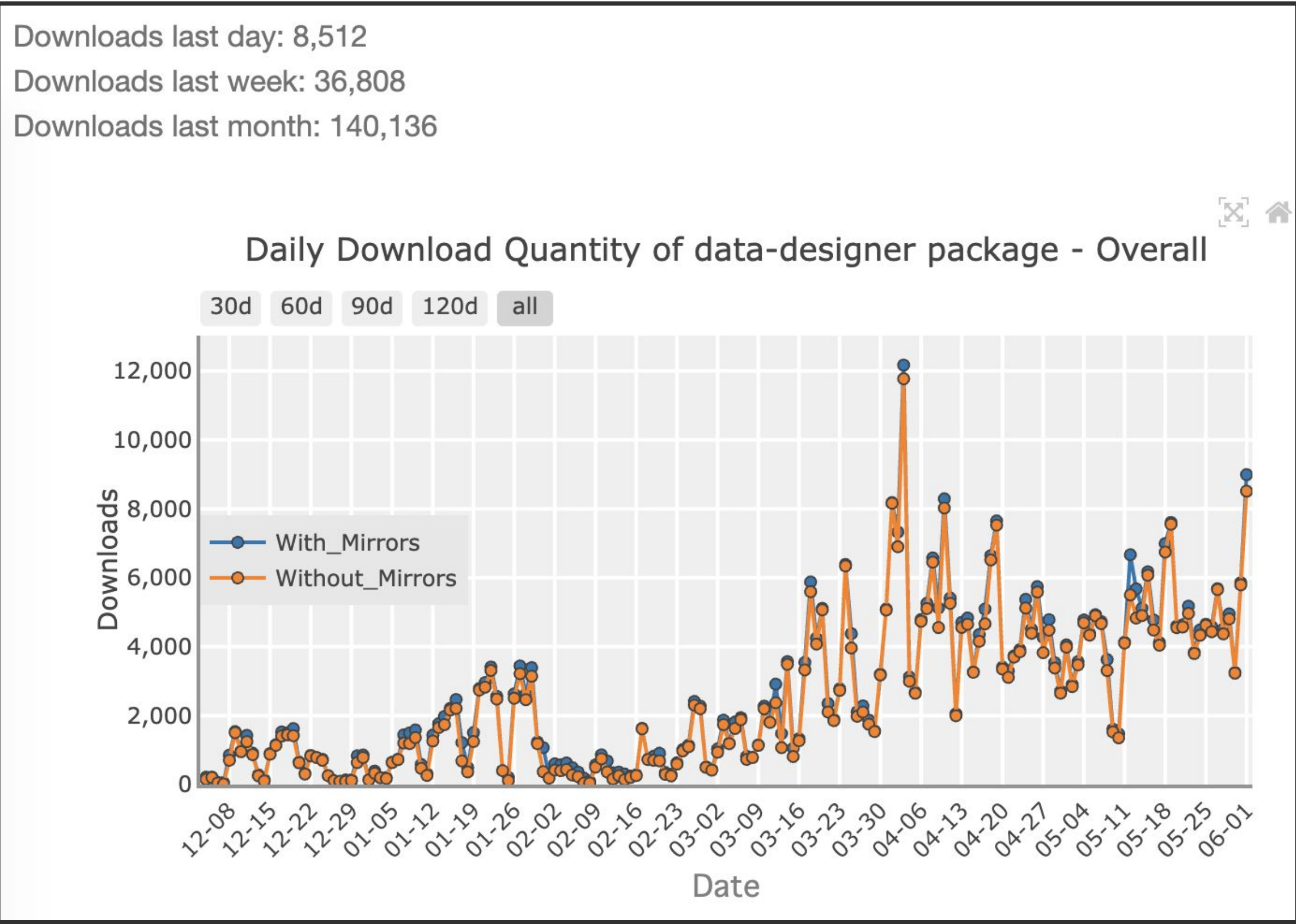
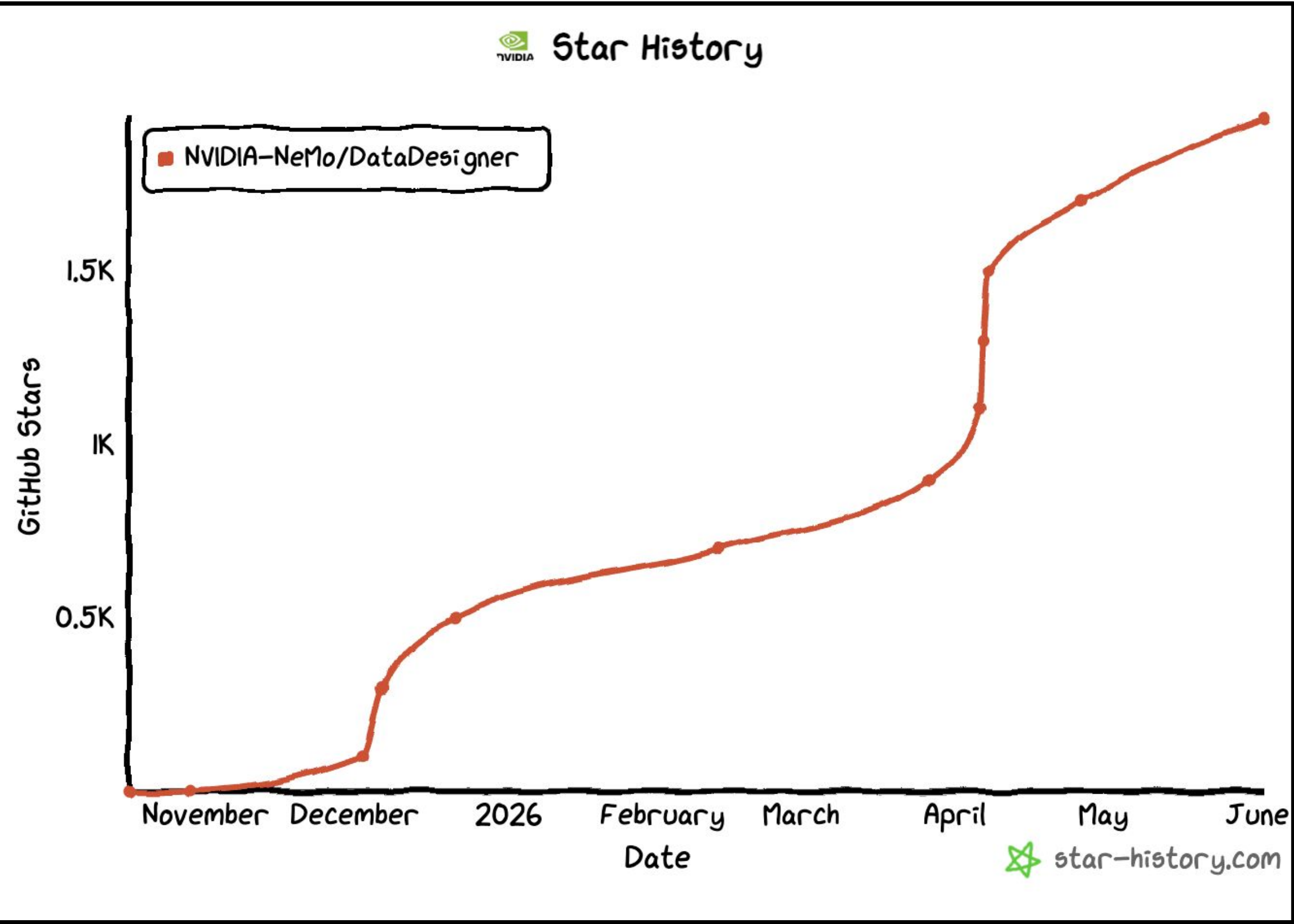
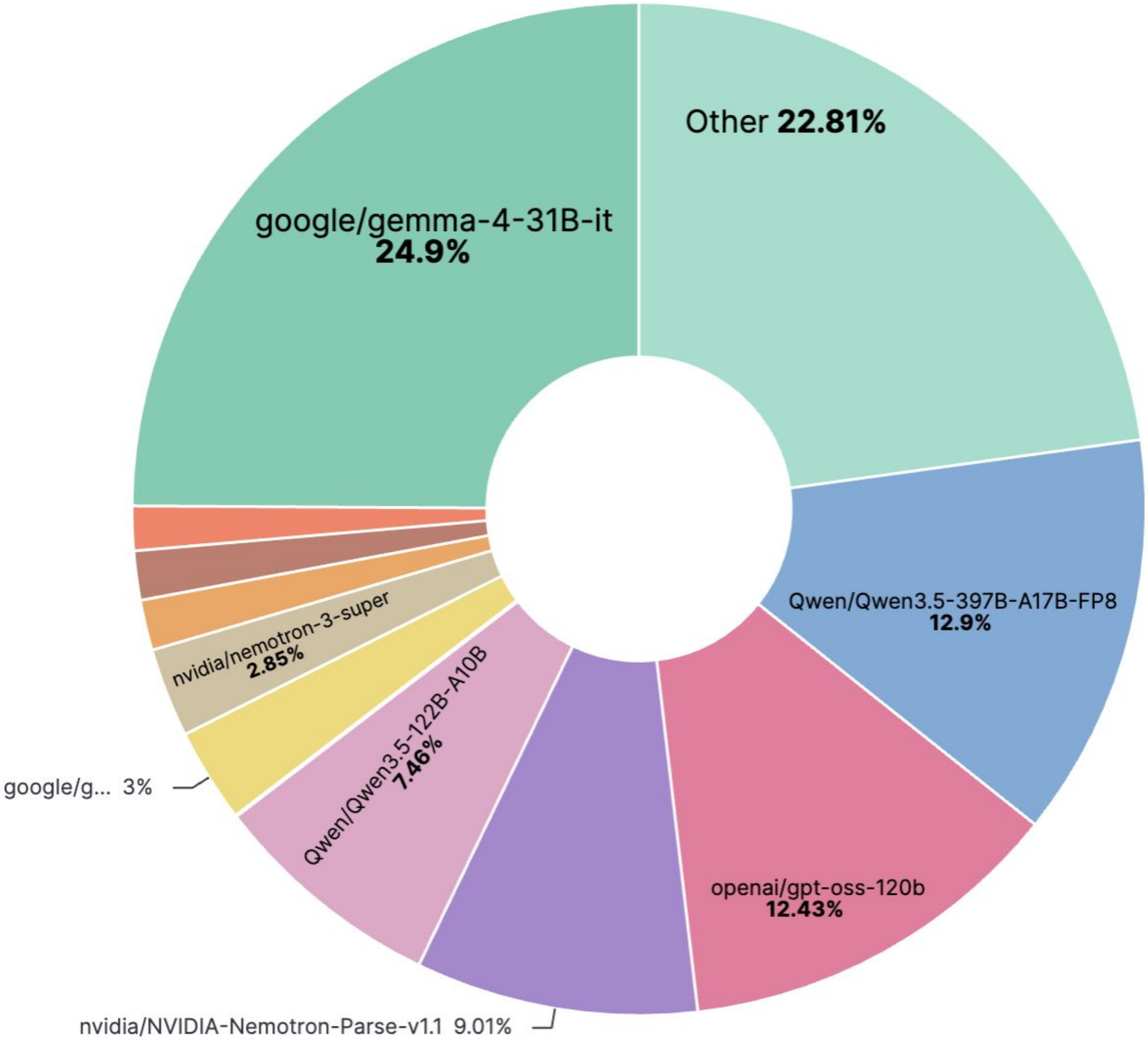
Context Tokens

562B+

Generated Tokens

1.2M+

SDG Jobs



NeMo Data Designer

Contributions to Nemotron Models

Goal	SDG Tokens	Results
Search Tool Calling Accuracy for Nemotron Super v3	7000 records, 0.087B tokens	0% to 31.28% on BrowseComp
Multi-Turn Tool Call Accuracy for Nemotron Nano, Super, Ultra v3	2M Records	2.5% to 27% on multi-turn accuracy
Structured Output Generation for Nemotron Nano, Super, Ultra v3	0.07B Tokens	+6.3% on JSONSchemaBench (80.2% → 86.5%) +8.0% across StructEval-T (TOML, JSON, YAML, CSV and XML)
Structured Query Language Performance Nemotron Nano, Super, Ultra v3	96.5k records	26.77% to 41.80% BIRD-bench
Cross Domain Scientific Reasoning (InfiniByte) for Nemotron Nano, Super, Ultra v3	14.9B Tokens	Validated SDG by showing a 10% improvement on Qwen-2.5 on SciCode Benchmark
STEM performance for Nemotron Nano, Super, Ultra v3 <i>This contributed to 2% of all pretraining data, making it the 3rd-highest-contributing SFT dataset in Super pretraining</i>	140B Tokens	• HumanEval Plus (+3% absolute) • MMLU-Pro (+4% absolute) • Math Reasoning (5% absolute) • MBPP (+3% absolute)
Nemotron-Personas A collection of multilingual synthetic persona datasets that support sovereign AI development across many countries and regions. Used to produce more diverse and rich data across variety of applications	Billions	• Safety data • Instruction-following data • Chat data • Multi-turn tool calling data
Long document understanding for Nemotron 3 Nano Omni	11.5M Records 45B Tokens	26% to 57.5% on MMLongBench-Doc

NeMo Data Designer

Technical posts from the core team + contributors

<https://docs.nvidia.com/nemo/datadesigner/dev-notes/overview>

NeMo Data Designer

Latest ▾

Ask AI ✨

NVIDIA-NeMo/DataDesigner
☆ 1943 🗨 178

Discover

Code Reference

Overview

Config ▾

Interface ▾

Engine Extension API ▾

Dev Notes

Overview

Designing Nemotron-Personas

Prompt Sensitivity

Retriever SDG Toolkit

Have It Your Way

VLM Long Document Understanding

Push Datasets to Hugging Face Hub

Text-to-SQL for Nemotron Super

Async All the Way Down

Dev Notes

Dev Notes

✨ Ask a question | 📄 Copy page | 📄 View as Markdown | More actions ▾

Welcome to NeMo Data Designer Dev Notes — in-depth guides, benchmark write-ups, and insights from the team building NeMo Data Designer.

Nemotron-Personas

JUN 1, 2026

Designing Nemotron-Personas

The compound-AI pipeline behind Nemotron-Personas seeding Nemotron training. Now open-sourced: fork it, add your data, ship

Prompt Sensitivity

Small changes in prompt.
Same intent. Same answer.

MAY 28, 2026

Mitigating Prompt Sensitivity

A Data Designer pipeline that generates thousands of regex-paired prompt preambles to manufacture format robustness —

Feedback

NeMo Data Designer

Why do we need it?



```
1 records = []
2
3 for _ in range(1_000_000):
4     records.extend(
5         [
6             {"customer": llm.generate("Generate a customer profile.")},
7             {"product": llm.generate("Generate a product that the customer would buy.")},
8             {"review": llm.generate("Generate a review for the product.")},
9         ]
10    )
11
12 synthetic_dataset = pd.DataFrame.from_records(records)
```

Is this Synthetic Data Generation?

NeMo Data Designer

Why do we need it? What's missing?

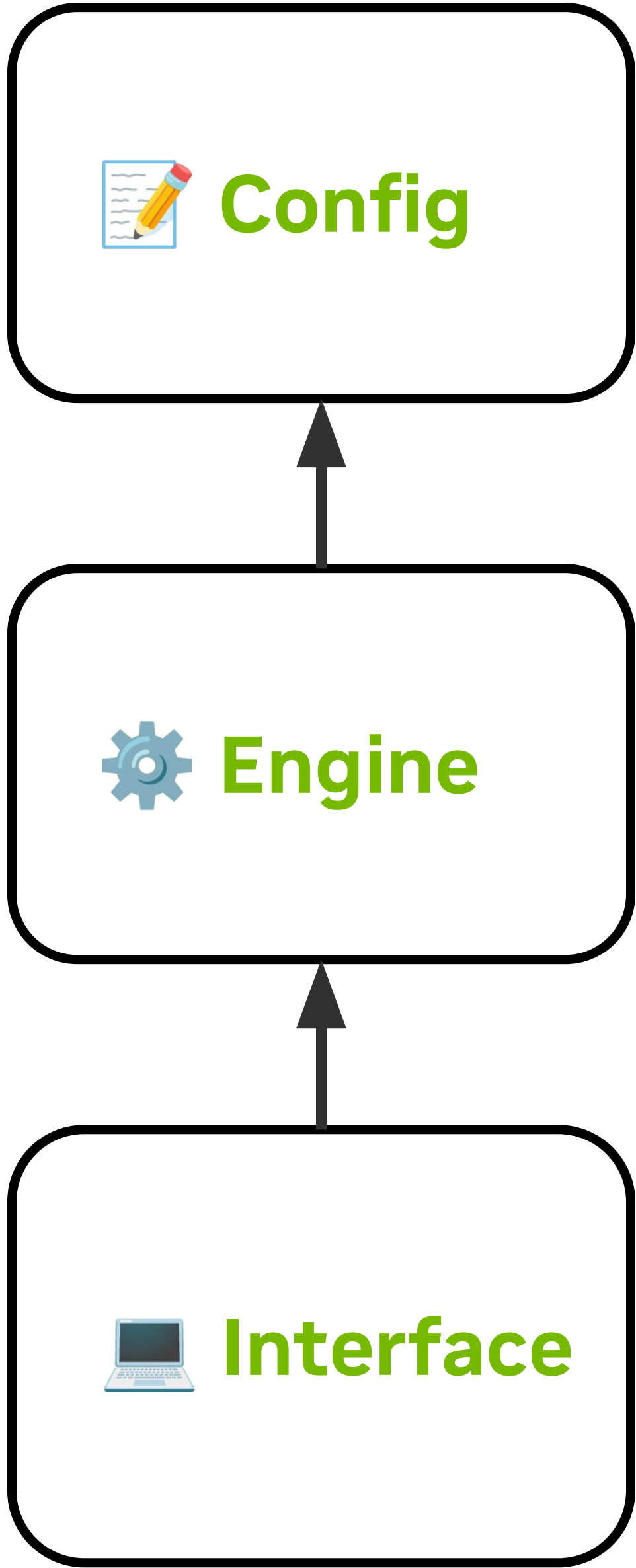
- Diversity, diversity, diversity!
- Steerability
- Correlations
- Validation / verification
- Reproducibility
- Scalability
- Multimodality



```
1 records = []
2
3 for _ in range(1_000_000):
4     records.extend(
5         [
6             {"customer": llm.generate("Generate a customer profile.")},
7             {"product": llm.generate("Generate a product that the customer would buy.")},
8             {"review": llm.generate("Generate a review for the product.")},
9         ]
10    )
11
12 synthetic_dataset = pd.DataFrame.from_records(records)
```


NeMo Data Designer

How does it work?



Column Types

Seed

Sampler

LLM Text

LLM Structured

LLM Code

LLM Judge

Validation

Expression

Custom Plugin

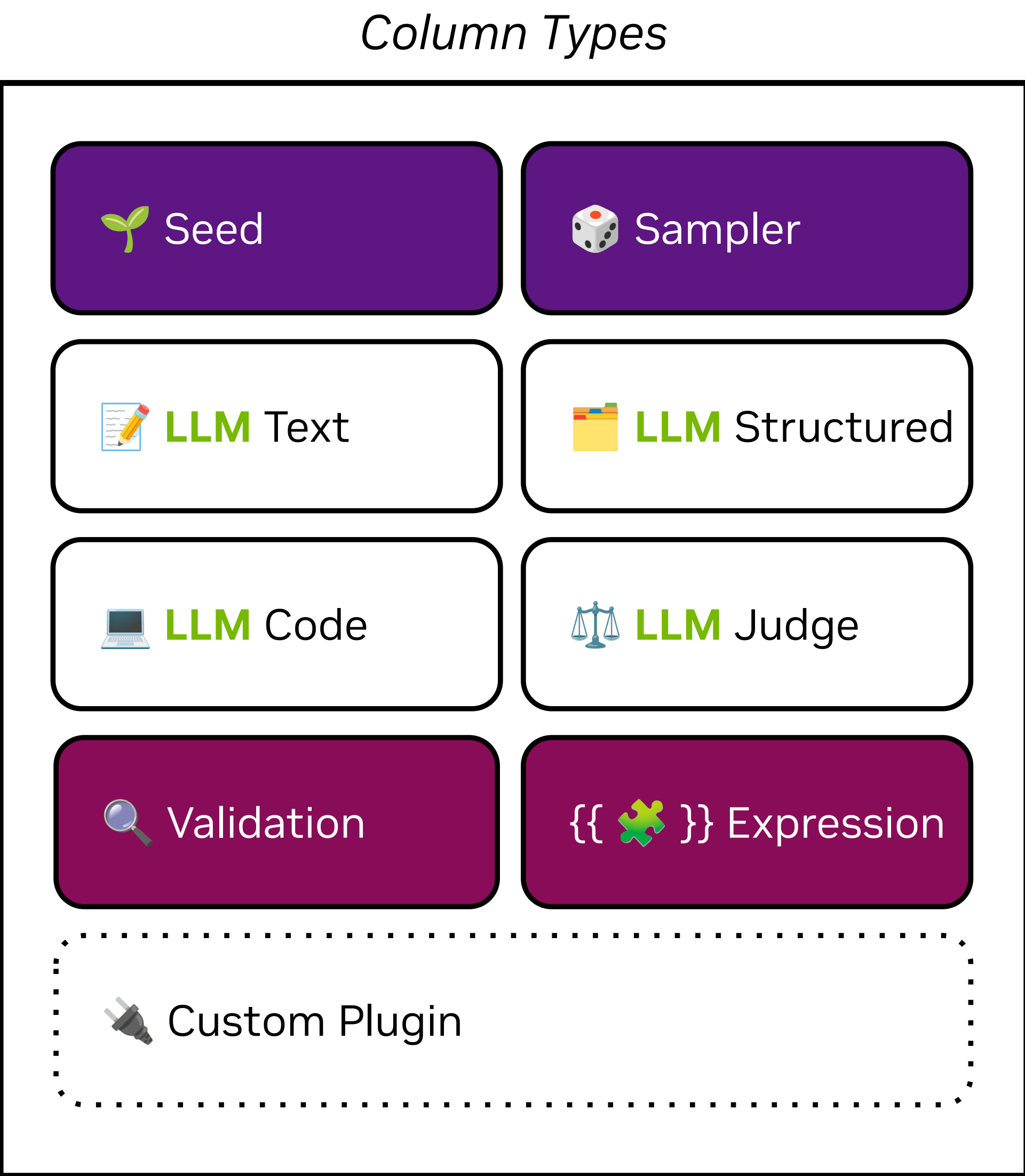
docstrings	coder_style	code	validation_result
...	...		...
...	...	...	...

```
1 config_builder.with_seed_dataset(Seed: docstrings)
2
3
4 config_builder.add_column(Sampler: coder style)
5
6
7 config_builder.add_column(LLM Code: Python functions)
8
9
10 config_builder.add_column(Validation: Code validator)
11
12 ...
```

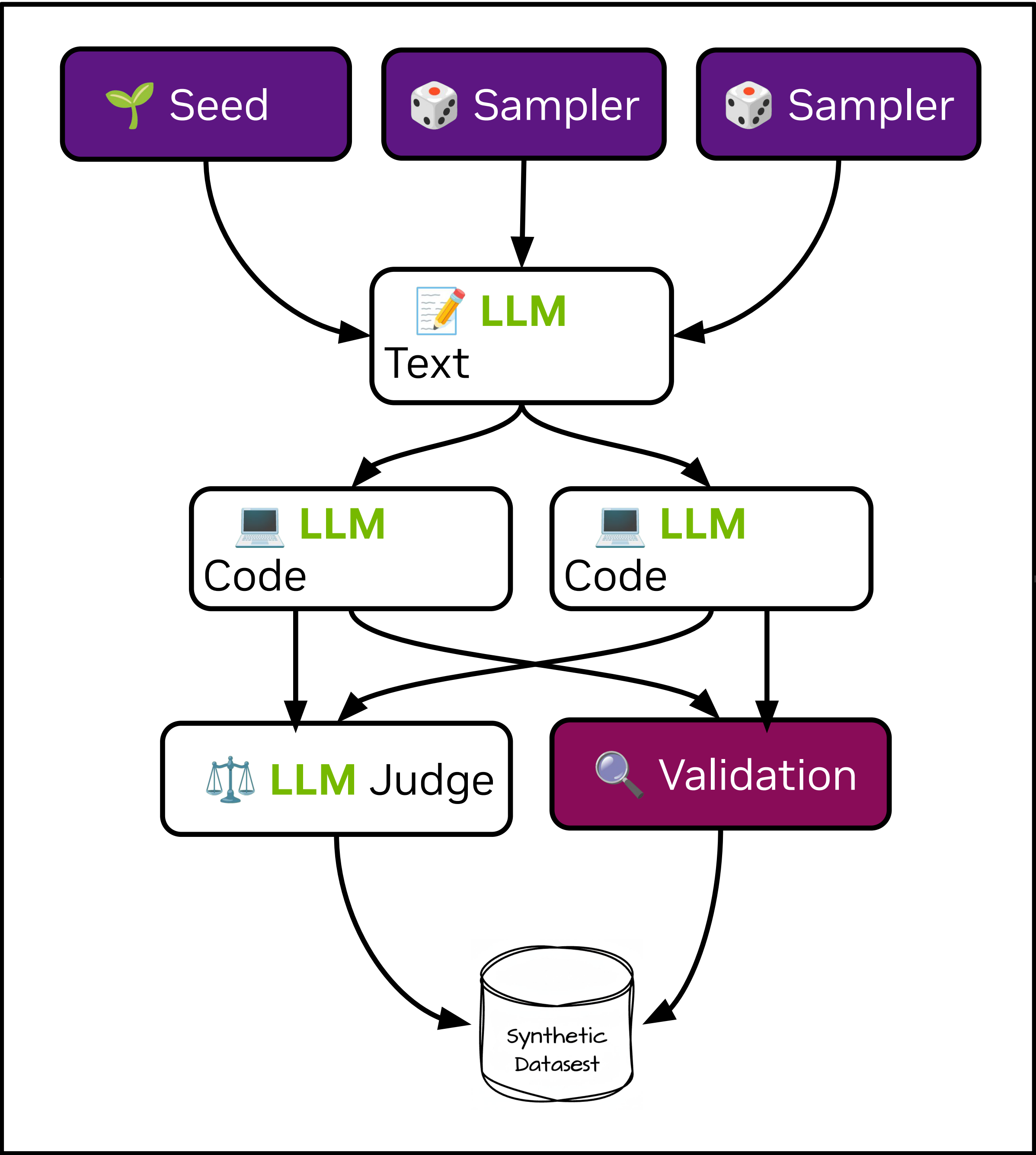
Design Dataset

NeMo Data Designer

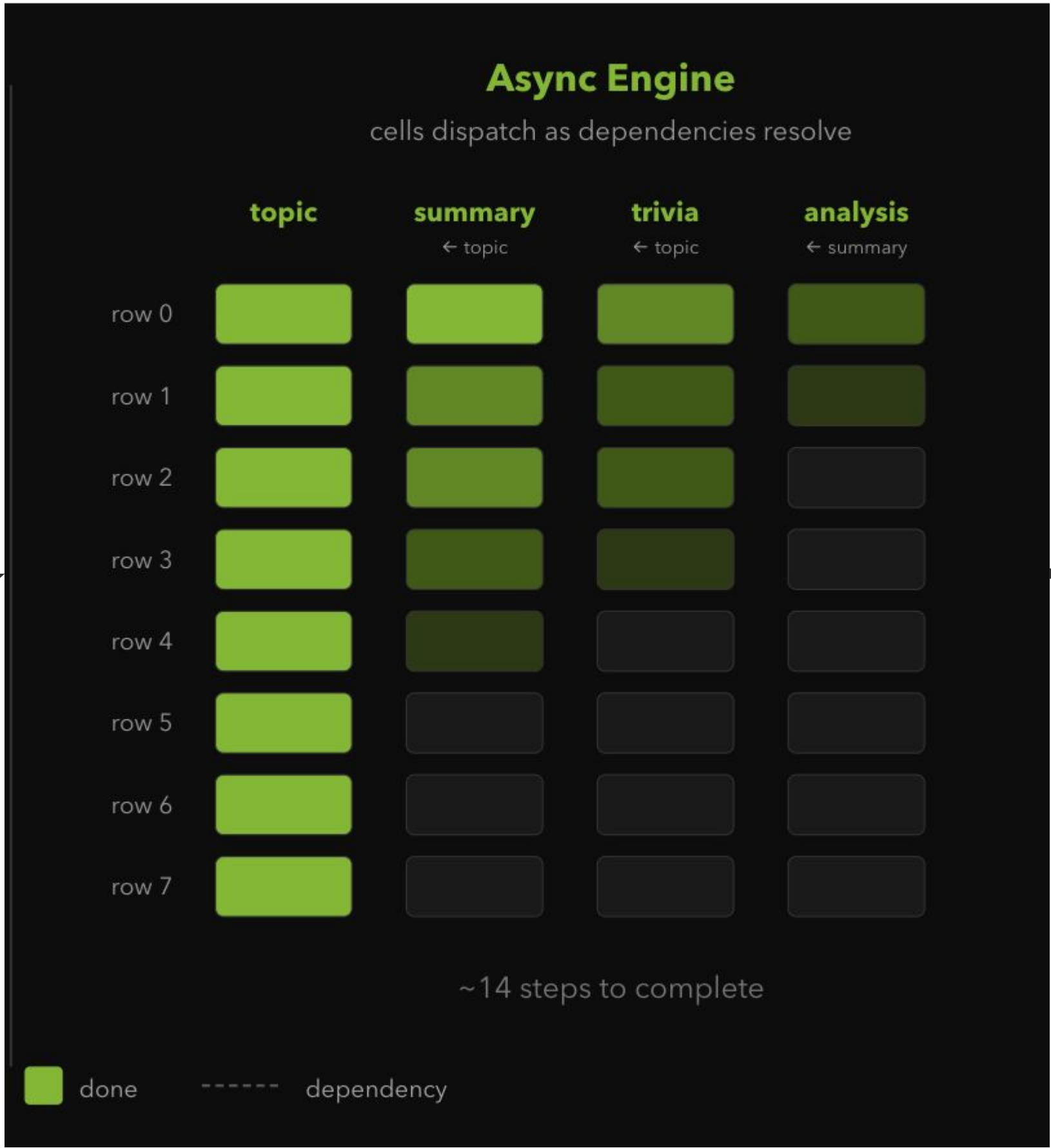
How does it work?



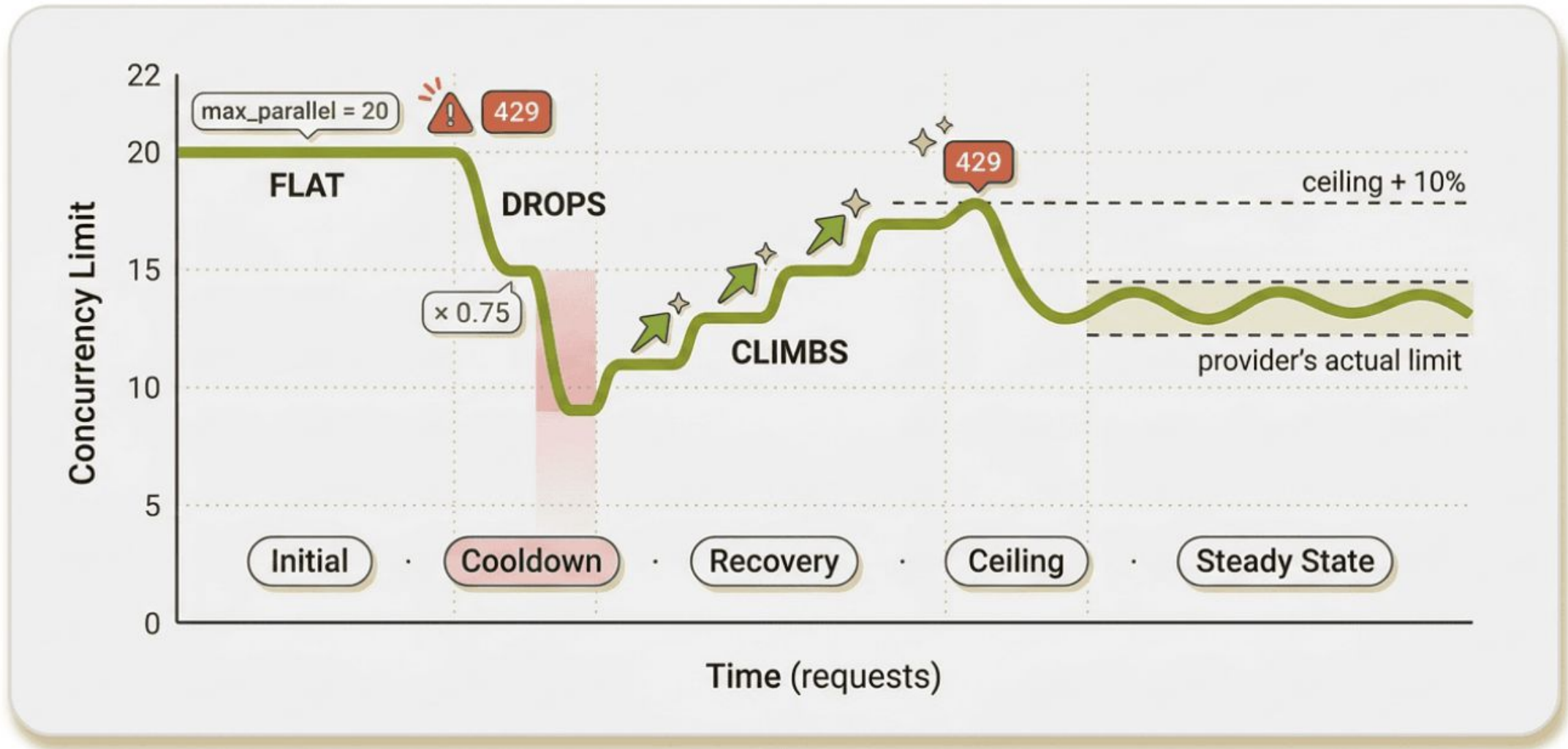
Create Dataset



Dataset Builder workflow



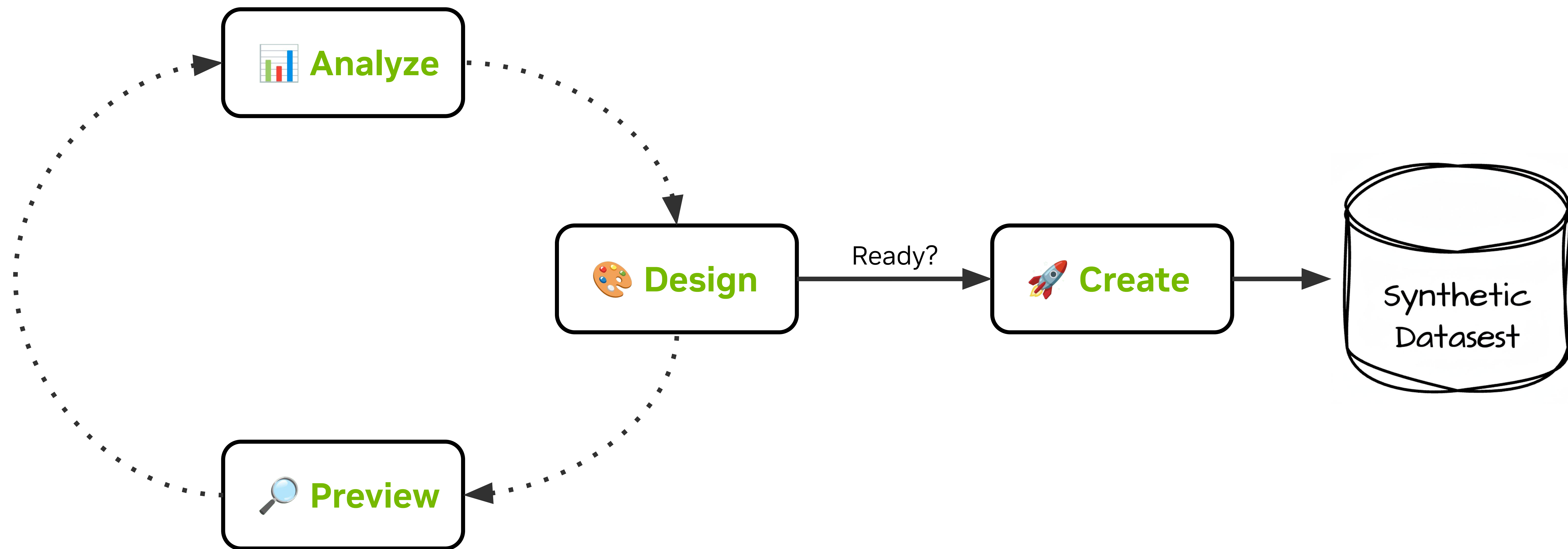
Cell Level Async Task Dispatch



AIMD Concurrency Control

NeMo Data Designer

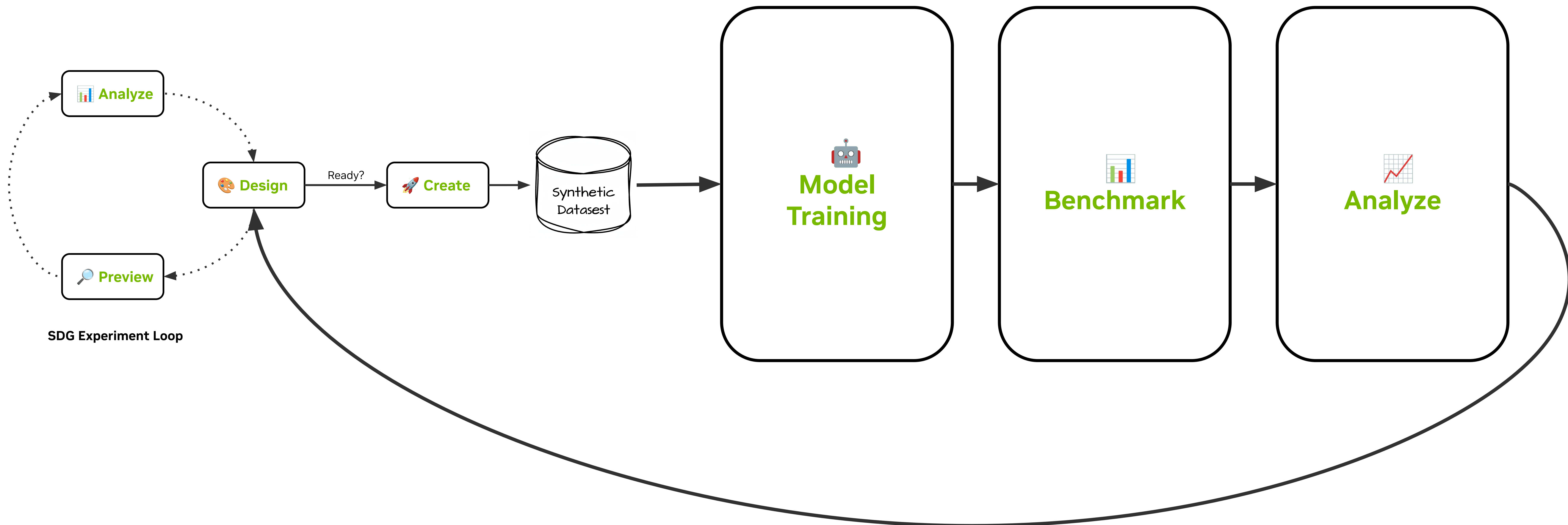
Experiment lifecycle



SDG Experimentation Loop

NeMo Data Designer

Experiment lifecycle



Model Training Lifecycle with SDG

NeMo Anonymizer



NeMo Anonymizer

What is it?

A text anonymization library built for the LLM era.

Transform the minimal set of details needed to **break sensitive inference while preserving useful meaning**.

Open Source and built on NeMo Data Designer

<https://github.com/NVIDIA-NeMo/Anonymizer>

```
pip install nemo-anonymizer
```

CLASSIC METHODS ARE INSUFFICIENT

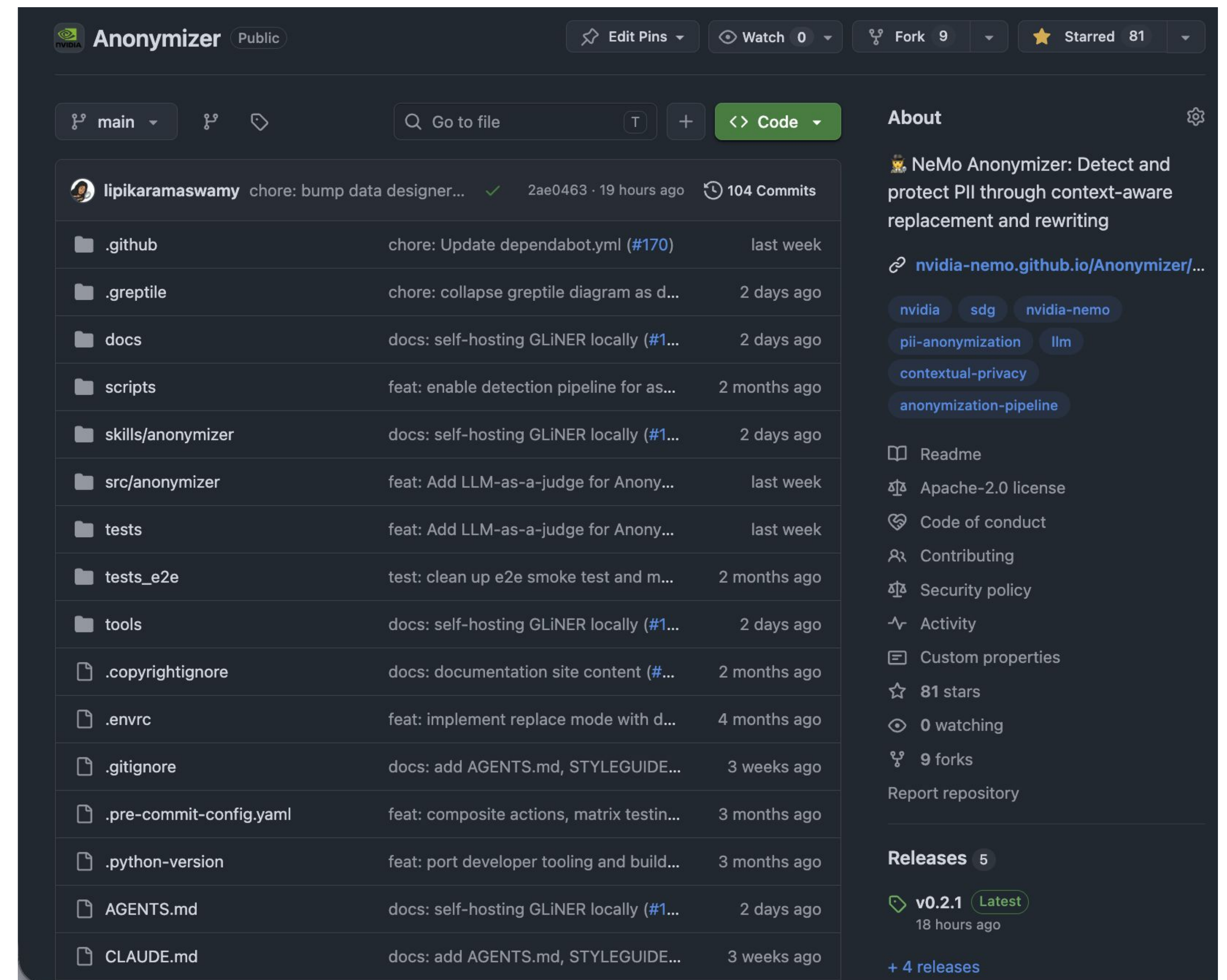
"Sarah described her appointment. She's looking forward to ringing the bell soon and said the care team has been wonderful."

Sarah → first name

Standard PII detection identifies the explicit identifier.

Ringing the bell → reveals cancer treatment

Latent contextual clues reveal a sensitive medical status that traditional tools miss.



NeMo Anonymizer

Why do we need it?

Modern LLMs can reason over 'clean' text — re-identifying without ever seeing a name.

Quasi-identifiers do the damage

- Gender + DOB + ZIP uniquely identifies 63–87% of Americans — no name required
- Combinations matter more than any single field

Privacy is contextual, not categorical

- What's sensitive depends on surrounding text and your privacy goal — not a fixed PII list

Inference got cheap

- LLMs recover sensitive attributes from context, even after obvious PII is stripped
- Research confirms recovery via contextual reasoning alone

Agents have infinite tools

- Web search + public records + breach corpora = re-identification in a few tool calls

NeMo Anonymizer

How does it work?

Detect Sensitive Entities

A hybrid NER model + LLM pipeline identifies

- Explicit identifiers
- Latent contextual clues

Then apply one of

5 Transformation Options



Annotating

Adds identified entity to original PII



Redacting

Substitutes PII with a generic tag



Hashing

Substitutes PII with a hashed value



Substituting

Substitutes PII with a context-relevant alternative



Rewriting

Rewrites entire phrases while maintaining context

Notebook 1: Text QA SDG with NeMo Data Designer



Notebook 1: Text QA SDG with NeMo Data Designer

Find notebook [here](#) on github

Notebook 2: VQA with SDG with NeMo Data Designer



Notebook 2: VQA with SDG with NeMo Data Designer

Find notebook [here](#) on github

Notebook 3: Privacy with NeMo Anonymizer



Notebook 3: Privacy with NeMo Anonymizer

Find notebook [here](#) on github

Resources



NeMo Open-Source Data Libraries

From raw ingestion and cleaning to synthetic data generation and privacy protection

NeMo Curator

Data Curation

Clean, filter, and prepare safe multimodal data for agentic use cases and model training.

[Documentation](#)

NeMo Data Designer

Synthetic Data Generation

Create domain-specific synthetic datasets from scratch for building and evaluating specialized agents.

[Documentation](#)

NeMo Anonymizer

Data Anonymization

Perform context-aware data anonymization to protect PII while preserving insights.

[Documentation](#)

NeMo Safe Synthesizer

Synthesize Sensitive Data

Generate safe, synthetic versions of your sensitive datasets with no one-to-one mapping to original records.

[Documentation](#)